Quantitative Integration

HN-FE01 | Final examination | 100 points | 120 minutes

Calculate a coherent plan while preserving quantities, assumptions, and case identity.

Name _____ Date _____

Answer all eight questions. Q1-4 carry 10 points each; Q5-8 carry 15 points each. Calculator permitted. Carry full precision until display. All cases are fictional. Use extra paper as needed; label each answer by question and part. No physical test is required.

Question 1 Resting expenditure models

HN-FE01-Q01 | 10 points

Case QUANT-A supplies body mass 84 kg, height 178 cm, age 29 years, estimated body fat 25%, and MSJ branch constant+5. Use MSJ=10W+6.25H-5A+s. Optional equations are 500+22F and 370+21.6F.

a) [4 points] Calculate MSJ predicted resting expenditure.

b) [3 points] Calculate estimated FFM and both optional equation outputs.

c) [3 points] Explain why these outputs do not identify a measured true TDEE or automatically rank equation accuracy.

Question 2 Energy and macro allocation

HN-FE01-Q02 | 10 points

Continue QUANT-A. Use the selected MSJ output, inclusive multiplier 1.60, planned intake adjustment-15%, protein 2.0 g/kg BW, fat 0.75 g/kg BW, and 4/9/4 kcal per gram.

a) [4 points] Calculate the TDEE scenario and planned intake without internal rounding.

b) [3 points] Calculate protein, fat, and carbohydrate grams/day and check energy closure.

c) [3 points] State two assumptions that a dashboard should retain.

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Question 3 Activity accounting

HN-FE01-Q03 | 10 points

An assigned 6-MET activity lasts 35 minutes for the 84 kg case. Use gross MET×3.5×kg/200×minutes and the same expression with MET-1 for the incremental amount above the assigned 1-MET reference.

a) [4 points] Calculate gross energy.

b) [3 points] Calculate incremental energy and the reference component included in gross.

c) [3 points] May the gross value be added to the already inclusive TDEE scenario without redefining the boundary?

Question 4 Load estimates and session time

HN-FE01-Q04 | 10 points

A comparable log records 90 kg for 6 entered repetitions. Use Epley-style estimated 1RM=L(1+r/30). A work block has four 45-second sets and 150-second rests only between adjacent sets.

a) [4 points] Calculate estimated 1RM and distinguish estimate from measurement.

b) [3 points] Calculate 75% of that estimate and round to the nearest 2.5 kg total increment.

c) [3 points] Calculate the defined work-block duration.

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Question 5 Volume and progression audit

HN-FE01-Q05 | 15 points

A target has D=4 direct sets and I=8 assigned indirect sets. A separate exercise uses 90 kg, three sets, range 8-12, minimum RIR 2, total increment 2.5 kg. Two distinct comparable exposures each have repetitions 12, 12, 12 and RIR 2, 2, 2; no concern.

a) [5 points] Calculate direct-only, fractional(alpha 0.5), and total-contribution counts.

b) [5 points] Apply the course double-progression policy and calculate percentage load change.

c) [5 points] Change only the current final set to 11 repetitions. Give the decision and explain why physical sets and per-muscle counts must remain distinct.

Question 6 External and internal moments

HN-FE01-Q06 | 15 points

An abstract pivot has force 120 N, radius 0.40 m, and perpendicular pivot-to-force-line distance 0.25 m. For a separate explicitly static single-balancing-muscle model, the internal moment arm is 0.03 m.

a) [5 points] Calculate external moment magnitude and identify the correct distance.

b) [5 points] Under the stated single-muscle equilibrium assumption, calculate balancing muscle force.

c) [5 points] Explain two reasons the answer is not an in-vivo tissue-force or injury-risk measurement.

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Question 7 Trend and expenditure proxy

HN-FE01-Q07 | 15 points

Complete adjacent seven-day means are 84.2 and 83.8 kg; centers are seven days apart. Assigned mean intake 2465 kcal/day. Use the optional static proxy intake-7700×(mean change/7).

a) [5 points] Calculate mean change and percentage relative to the first mean.

b) [5 points] Calculate the static expenditure proxy and compare numerically with the 2900 scenario.

c) [5 points] Explain why close numerical agreement would not establish true expenditure or exact fat loss.

Question 8 Repair an integrated report

HN-FE01-Q08 | 15 points

A report says: Every lift must increase each session. Add 308.7 gross cardio calories to inclusive TDEE. The observed 0.4 kg decline proves fat loss. A week 5 count 12 proves MRV. The dashboard export proves adherence.

a) [5 points] Correct the progression and energy-boundary claims.

b) [5 points] For a separate one-category base 9 six-week planner, use offsets 0, 1, 2, 2, 3 and week 6=ceil(week 5/2). Give the counts, total, and week 6 reduction.

c) [5 points] Correct the tissue-change, MRV, and adherence claims and propose one review action.